NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Pathology of the Lower Urinary Tract

Eugen Petcu, MD, PhD
Associate Professor,
Department of Biomedical Sciences
epetcu@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Session Objectives

At the end of this session, you will be able to

1. Discuss the etiopathogenesis and morphology of **congenital anomalies** of the ureter and the bladder
2. Describe the etiopathogenesis and morphology of **infectious, reactive and inflammatory lesions** of the ureter, bladder and urethra
3. To become familiar with the etiopathogenesis and morphology of **neoplastic lesions** of the ureter, bladder and urethra.

Please note that unless mentioned otherwise, all the images and information included in this presentation are from Robbins and Cotran, Pathologic Basis of Disease, 10th edition

Ureters: Congenital Anomalies

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

VUR Reflux
Double and Bifid Ureters
Ureteropelvic junction obstruction

These are seen in 2% to 3% of all autopsies. Most have little clinical significance;

Vesicoureteral reflux (VUR)

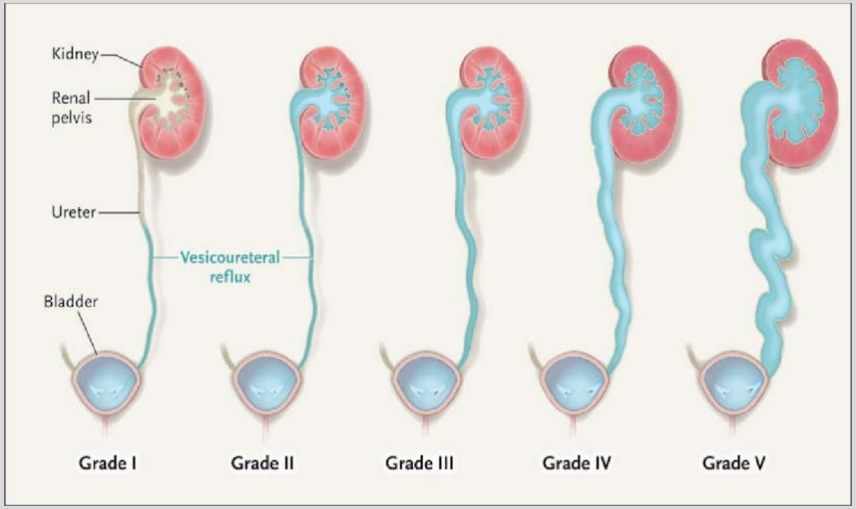
It represents retrograde urine flow from the urinary bladder to the upper urinary tract.
Induces severe nephropathy. Early treatment is very important.
Voiding cystourethrogram (VCU) is the gold standard method of diagnosis

Pathogenesis: Steps

The ureter runs downwards and drains urine from the kidney into the urinary bladder.
Intramural/intravesical ureter: an oblique tunnel within the bladder.
Its orifice typically remains closed due to its obliquity.
The ureteric peristalsis increases the ureteric pressure.
Subsequently, the orifice opens to allow the passage of urine into the bladder.
Then the ureteric pressure drops while the intravesical pressure rises again and closes the intramural ureteral part.

The distal ureteric orifice remains open if the ratio of intramural tunnel length to the ureteral diameter is less than 5.
Also seen in immature bladder discoordination/ bladder dysfunction.

The international grading system of VUR



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

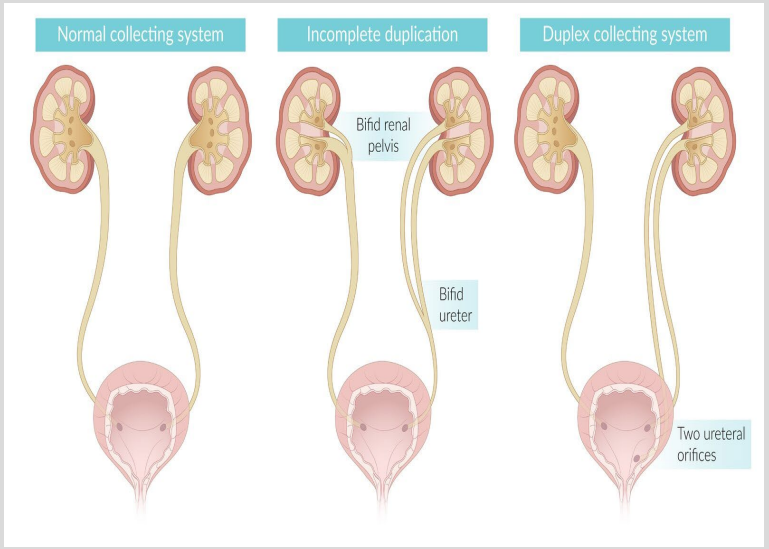
Grade 1: Reflux only into the non-dilated ureter.
Grade 2: Reflux into the ureter and the renal pelvis without dilatation.
Grade 3: Reflux with mildly dilated ureter and pyelocalyceal system.
Grade 4: Reflux with the tortuous and moderately dilated ureter with blunting of renal fornices. The papillary impression is preserved.
Grade 5: Reflux with the tortuous and severely dilated ureter, dilatation of pyelocalyces with loss of fornices, and papillary impression.

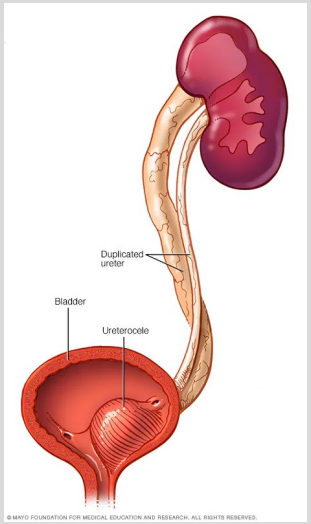
Double/Bifid Ureter

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Ureteral duplication: two ureters coming from the same kidney. There are two types:

- Cases with two entirely separate ureters coming from the same kidney (complete duplication).
- Cases with two ureters that join together again before they get to the bladder (incomplete duplication-bifid ureter).





Pathogenesis

In utero at the 4th week a ureteric bud, forms from the mesonephric (or Wolffian) duct, and determines formation of the ureter.

2 mechanisms:

- Double ureter: the ureteric bud either splits or there are two buds.
- The kidney is divided into two parts, an upper and lower lobe, with some overlap due to intermingling of collecting tubules and one ureter is formed. If the division is complete each kidney lobe has its own renal pelvis and ureter.

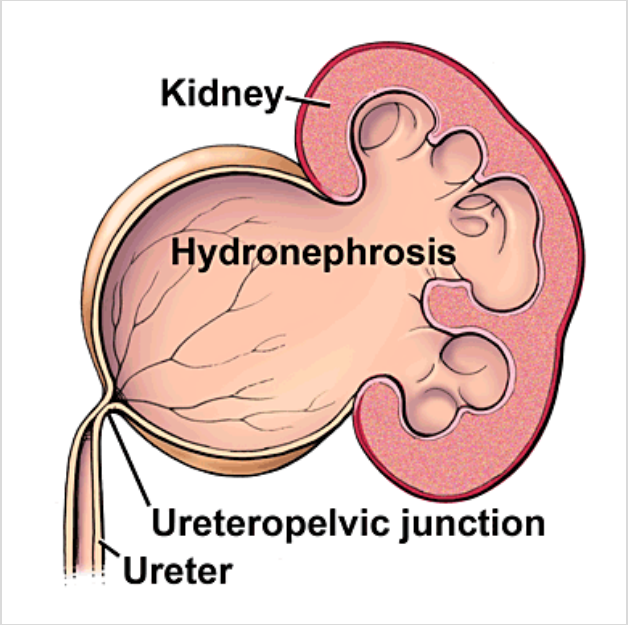
Ureteropelvic junction (UPJ) obstruction

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

It is the most common cause of hydronephrosis in infants and children which preferentially affects males and is bilateral in 20% of cases.
There is agenesis of the contralateral kidney in a minority of cases.

In adults: more common in women and is most often unilateral.

There is abnormal organization of smooth muscle bundles or excess stromal deposition of collagen between smooth muscle bundles at the UPJ obstruction.
Other cases are determined by extrinsic compression of the UPJ by abnormal renal vessels.



Ureter: Inflammatory conditions

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Ureteritis follicularis: Long standing cases, accumulation of lymphocytes in subepithelial region causes slight elevation of the mucosa

Ureteritis cystica: cysts on the mucosa.
It is asymptomatic.
It results from long-standing chronic inflammation.
Nests of transitional epithelium grow downward into the lamina propria and a central cystic space forms



Ureteritis cystica: opened ureter with multiple small glistening "bumps" on the surface.

Ureters: Obstruction

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Major Causes of Ureteral Obstruction

Cause of Obstruction	Characteristics/Mechanisms
Intrinsic	
Calculi	Typically of renal origin
	Usually small (5 mm in diameter or less)
	Impact at loci of ureteral narrowing—ureteropelvic junction, where ureters cross iliac vessels and where they enter bladder—and cause excruciating "renal colic"
Strictures	May be congenital or acquired
Tumors	Urothelial carcinomas arising in ureters
	Rarely, benign tumors or fibroepithelial polyps
Blood clots	Massive hematuria from renal calculi, tumors, or papillary necrosis
Neurogenic	Interruption of the neural pathways to the bladder
Extrinsic	
Pregnancy	Physiologic relaxation of smooth muscle or pressure on ureters at pelvic brim from enlarging fundus
Periureteral inflammation	Salpingitis, diverticulitis, peritonitis, sclerosing retroperitoneal fibrosis
Endometriosis	With pelvic lesions associated with scarring
Tumors	Cancers of the rectum, bladder, prostate, ovaries, uterus, cervix; lymphomas, sarcomas

Intrinsic and **extrinsic** lesions may obstruct the ureters and may give rise to hydroureter, hydronephrosis, and pyelonephritis.

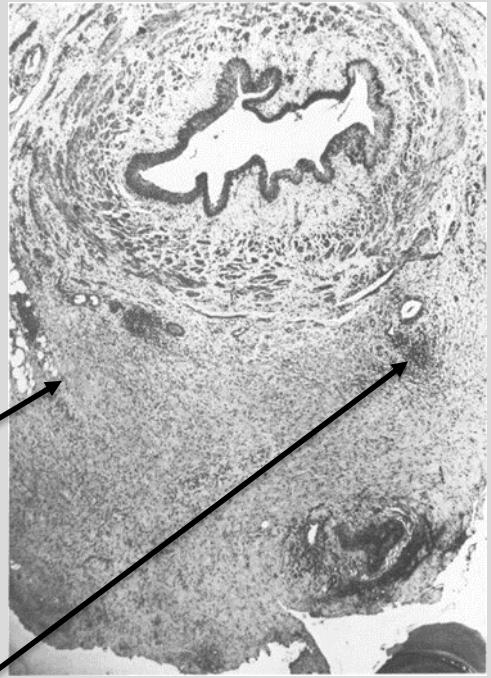
Sclerosing Retroperitoneal Fibrosis

NEW YORK INSTITUTE

It is a fibrotic inflammatory process that encases retroperitoneal structures and causes hydronephrosis.
-middle to late age and is more common in males than females.

It is associated with drug exposures (ergot derivatives, β-adrenergic blockers), inflammatory conditions (vasculitis, diverticulitis, Crohn disease), or malignancies (lymphomas, urinary tract carcinomas).
Most, however, have no obvious cause and are considered primary or idiopathic SRF.

Microscopic examination: fibrous tissue containing a prominent infiltrate of lymphocytes, often with germinal centers, plasma cells and eosinophils.



PMC full text:
Ann Surg. 1972 Aug; 176(2): 199–204.
doi: 10.1097/00000658-197208000-00013

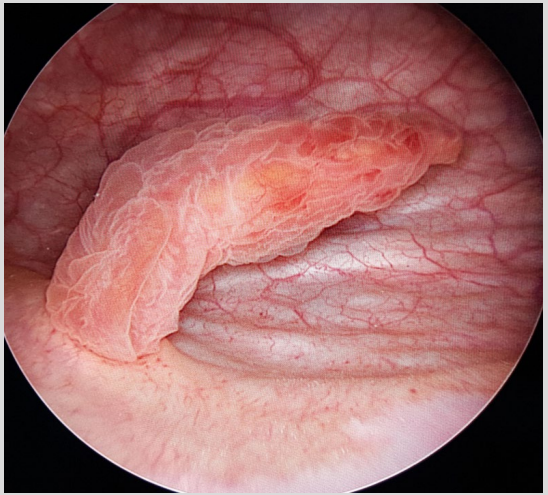
Ureters: Benign tumors

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

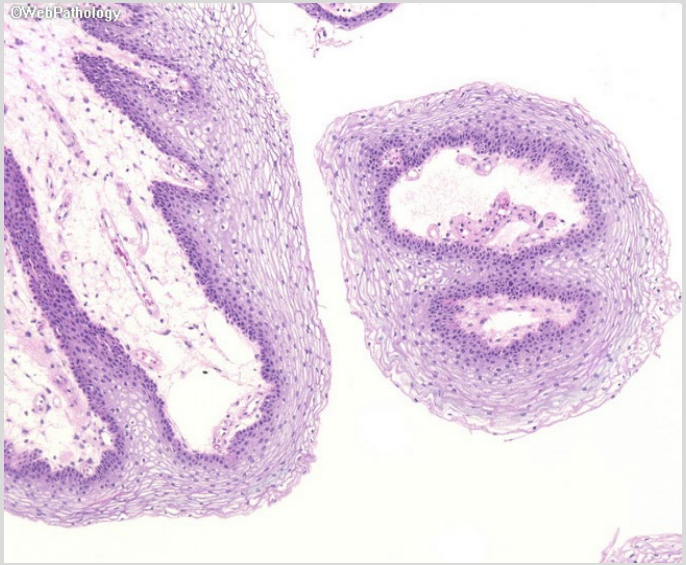
Primary tumors of the ureter are rare.

Fibroepithelial polyp= a tumor-like lesion often seen in children. Made of loose, vascularized connective tissue overlaid by urothelium.



A. Fibroepithelial polyp of the ureter prolapsing into the bladder

auanet.org/education/auauniversity/education-products-and-resources/pathology-for-urologists/normal-histology-and-important-histo-anatomic-structures/renal-pelvis-and-ureter



B. Histology: The squamous epithelium covering the polypoid areas lacks cytologic atypia.

https://www.eurorad.org/case/15230

Ureter: Malignant tumors

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Secondary malignant tumors of the ureter are metastases of renal and bladder primaries.

Primary malignant tumors of the ureter are similar with those arising in the renal pelvis, calyces, and bladder.
The majority are urothelial carcinomas.
They occur most frequently during the sixth and seventh decades of life and cause obstruction of the ureteral lumen.
They commonly occur concurrently with urothelial carcinomas of bladder or renal pelvis.



Papillary transitional cell carcinoma extensively involving the ureter.
(Courtesy Dr. Cristina Magi-Galluzzi, The Johns Hopkins Hospital, Baltimore, MD.)

Urinary Bladder: Congenital Anomalies

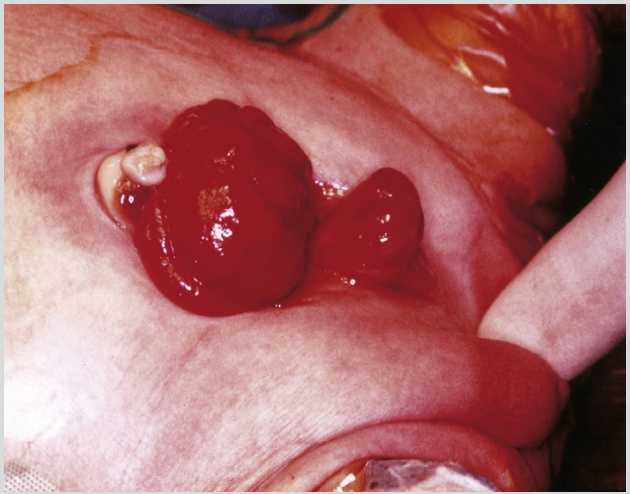
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Diverticula: 1 cm to 10 cm pouch-like invaginations of the bladder wall. Represent a risk for infection and bladder calculi.

- **Congenital diverticula** are caused by a focal failure of development of the normal musculature or to a urinary tract obstruction during fetal development.
- **Acquired diverticula** are associated with prostatic hyperplasia, producing urinary outflow obstruction. The resulting increase in intravesical pressure causes outpouching of the bladder wall and the formation of diverticula.

Exstrophy of the bladder

The bladder communicates directly with the abdominal surface. The exposed bladder mucosa may undergo glandular metaplasia and is subject to chronic infection that often spreads to the upper urinary tract



Exstrophy of the bladder in a newborn boy. The tied umbilical cord is seen above the hyperemic mucosa of the everted bladder. Below is an incompletely formed penis.
(Courtesy Dr. John Gearhart, The Johns Hopkins Hospital, Baltimore, MD.)

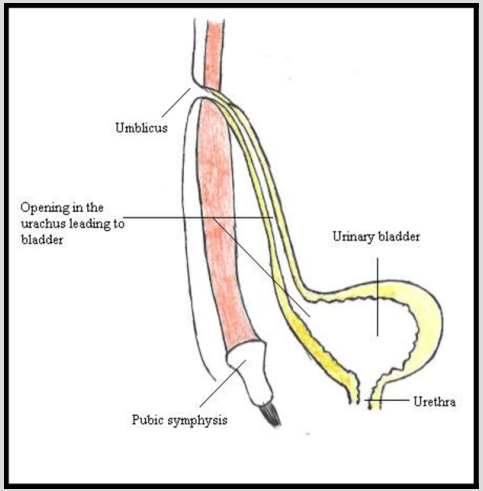
Urachal anomalies

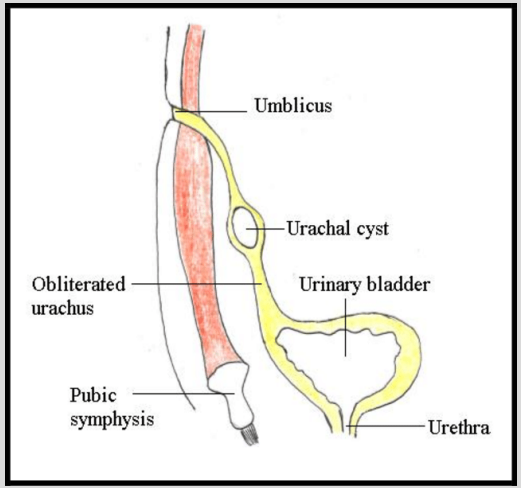
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

The urachal canal connects the fetal bladder with the allantois.
It is obliterated at birth.
When fully patent it creates a **bladder-umbilicus fistula**.

When only a central region of patent urachal canal persists, a **urachal cyst** lined by urothelial or metaplastic glandular epithelium is formed. Urachal cysts are at increased risk for neoplastic transformation, mostly presenting as adenocarcinomas-AC





https://www.researchgate.net/publication/313863179_CONGENITAL_ANOMALIES_OF_URACHUS_EMBRYOLOGICAL_BASIS_AND_ITS_CLINICAL_SIGNIFICANC

Urinary Bladder: Inflammation
Acute and Chronic Cystitis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Bacterial pyelonephritis is frequently preceded by infection of the urinary bladder.

Predisposing factors:
bladder calculi, urinary obstruction, diabetes mellitus, immune deficiency, instrumentation

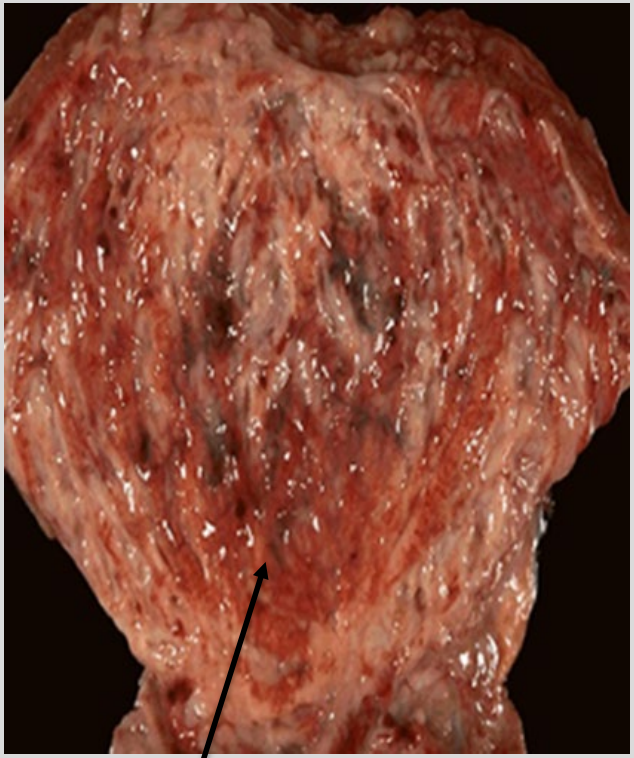
Etiology
Coliforms: Escherichia coli, Proteus, Klebsiella, Enterobacter.
Viruses (adenovirus), Mycoplasma.
Schistosomiasis (Schistosoma haematobium): cystitis in certain African and Middle Eastern countries.
Tuberculous cystitis after renal tuberculosis.

Women are more likely to develop cystitis as a result of their shorter urethras

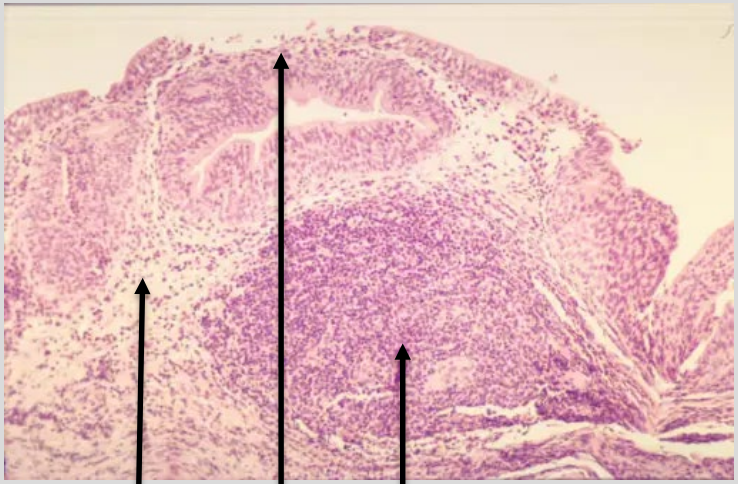
Acute Cystitis Morphology

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

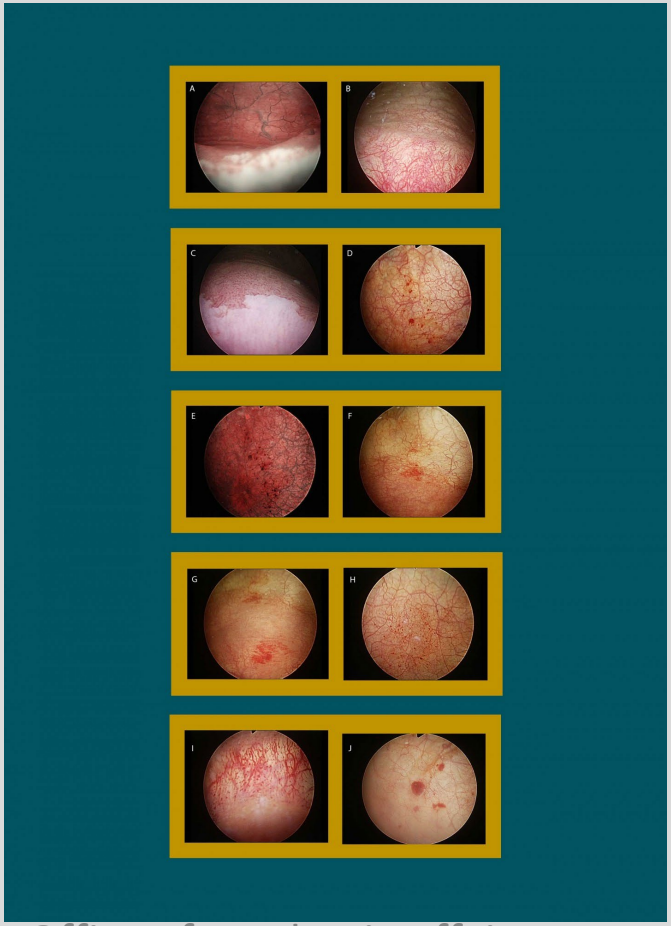


Hyperemia with small hemorrhagic areas



Histology: urothelial ulceration, inflammatory infiltrate above a suburothelial lymphoid follicle.

Chronic Cystitis Morphology



NEW YORK INSTITUTE OF TECHNOLOGY

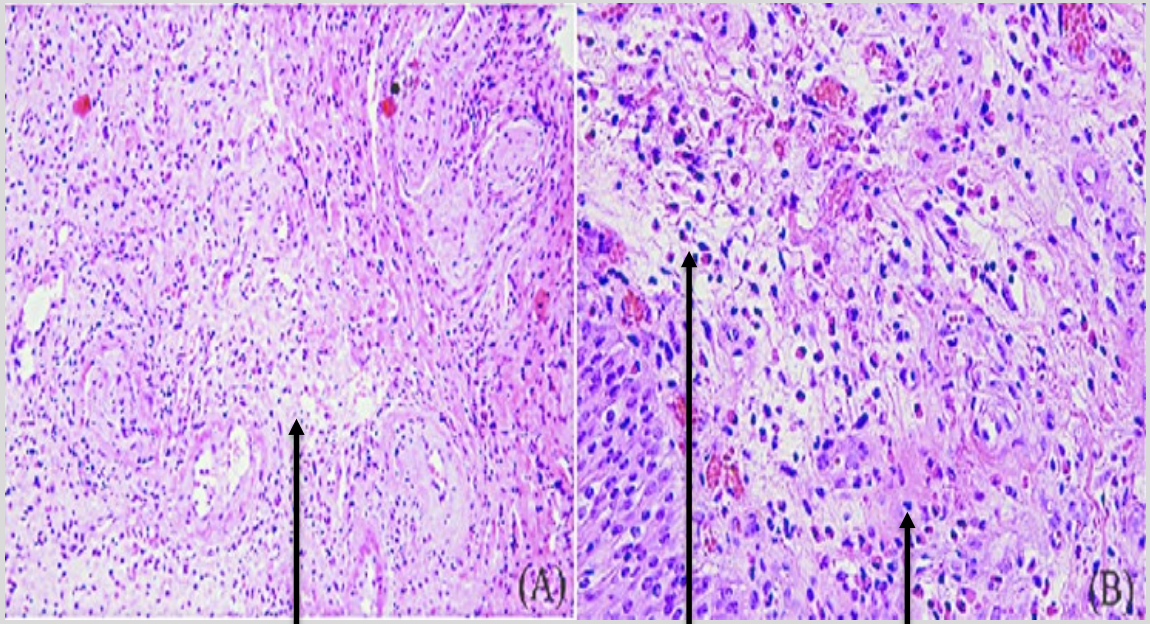
College of Osteopathic Medicine

Macro: friable, mucosa with some congestion and ulcerations.

https://www.georgiadis-urology.com/wp-content/uploads/2019/10/Different-types-of-Cystitis-Georgiadis-Urology.jpg

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Chronic Cystitis Histology



Histo: increased fibrosis and inflammatory cell infiltration, including many eosinophils, lymphocytes, and some histiocytes.

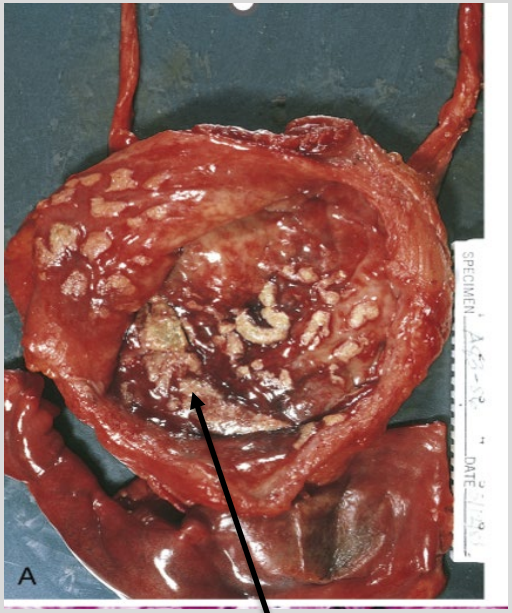
NEW YORK INSTITUTE OF TECHNOLOGY

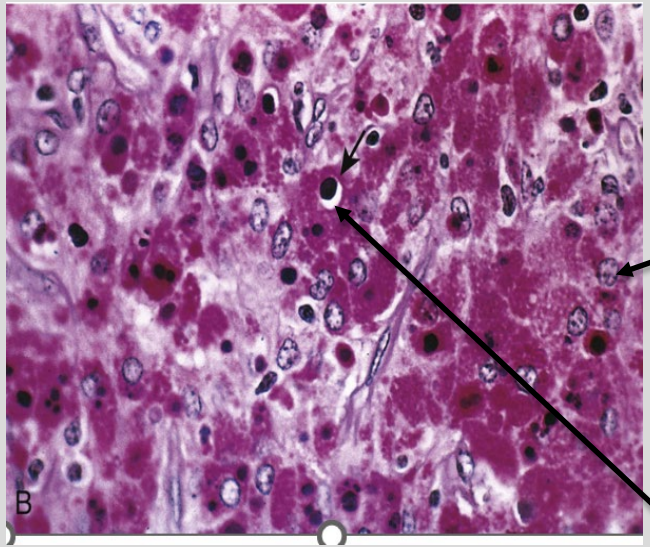
College of Osteopathic Medicine

Malakoplakia

It is distinctive chronic inflammatory reaction determined by acquired defects in phagocyte function.
It arises in the setting of chronic bacterial infection:E. coli or occasionally Proteus species (renal transplant recipients).

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine





Histo: large foamy macrophages , lymphocytes.
The macrophages have abundant granular cytoplasm due to phagosomes stuffed with debris of bacterial origin due to abnormal phagolysosome function.

Cystoscopically: soft yellow, slightly raised mucosal plaques, 3 to 4 cm in diameter

Laminated mineralized concretions resulting from deposition of calcium in enlarged lysosomes, known as Michaelis-Gutmann bodies, are typically present within the macrophages.

Obstruction of the bladder outlet

NEW YORK INSTITUTE OF TECHNOLOGY

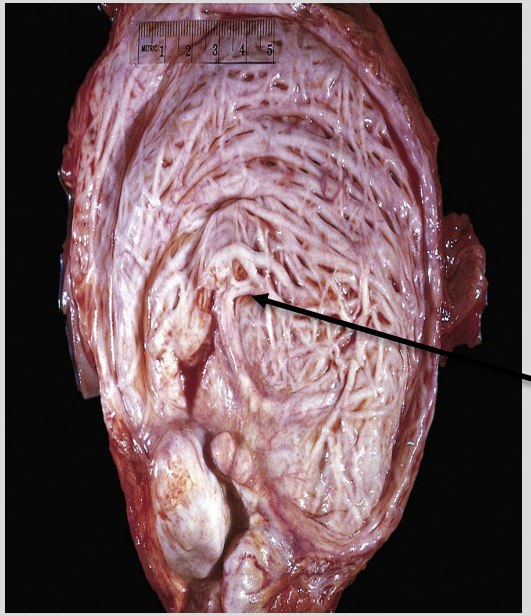
College of Osteopathic Medicine

It has a major clinical importance because of its eventual effect on the kidney.

Common in men; caused by BPH
Women: caused by cystocele of the bladder.

Other causes:

- congenital urethral strictures.
- inflammatory urethral strictures.
- bladder tumors, either benign or malignant.
- mechanical obstructions caused by foreign bodies and calculi.
- injury of nerves controlling bladder contraction (neurogenic bladder).



Morphology

Early: thickening of the bladder wall due to smooth muscle hypertrophy.

Long standing: the individual muscle bundles greatly enlarge and produce trabeculation of the bladder wall

Urothelial neoplasms

Tumors of the Urinary Bladder
Urothelial (transitional) tumors
• Noninvasive urothelial (transitional cell) tumors
• Infiltrating urothelial carcinoma
• Variants: nested, microcystic, micropapillary, plasmacytoid, sarcomatoid, giant cell, poorly differentiated, lipid-rich, and clear cell
Adenocarcinoma
Squamous cell carcinoma
Mixed carcinoma
Small-cell carcinoma
Sarcomas

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

They represent the ninth most common cancer type worldwide and the fourth most common cancer in American men (7% of all new cases). >95% of bladder tumors are of epithelial origin.

Urothelial Neoplasms Risk Factors

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Cigarette smoking
- the most important etiologic factor, increasing the risk 3-7X

Industrial exposure to aryl amines
-2-naphthylamine and related compounds
-Cancer appears 15 to 40 years after the first exposure.

Schistosoma haematobium infections
-endemic areas (Egypt, Sudan); the organism determines a chronic inflammatory response promoting a progressive mucosal dysplasia, and neoplasia.
-70% of these cancers are squamous, the remainder being urothelial or adenocarcinoma

Long-term use of analgesics
Heavy long-term exposure to cyclophosphamide
Irradiation

Urothelial Neoplasms Pathogenesis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Non–invasive papillary cancers

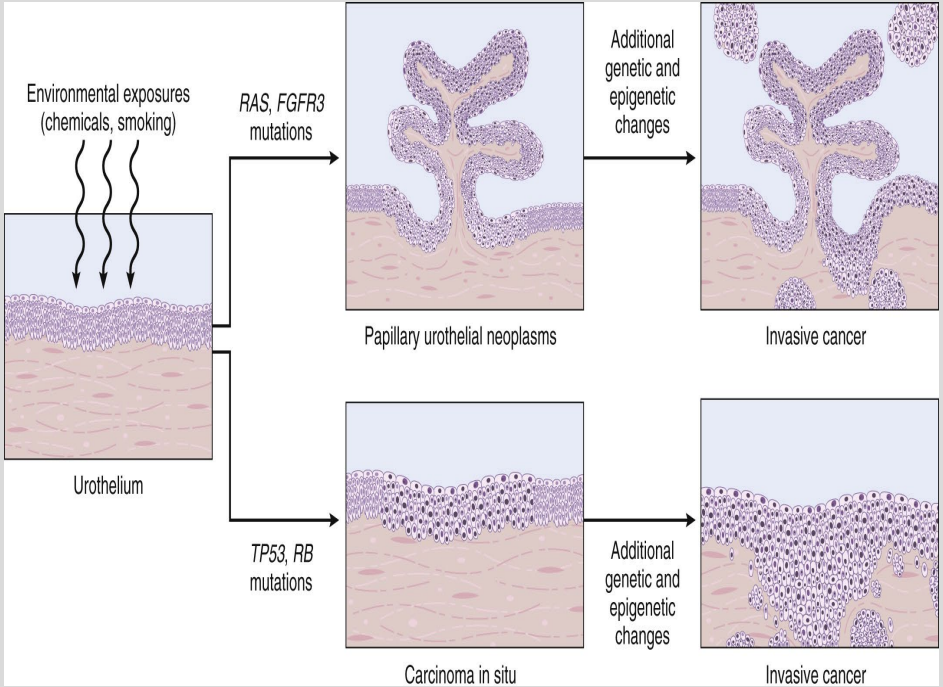
-Present with gain-of-function alterations that increase signaling through growth factor receptor pathway:

a. amplifications of the FGFR3 tyrosine kinase receptor gene and
b. activating mutations in the genes encoding RAS and PI 3-kinase.

-These tumors frequently recur but progress to muscle-invasive bladder cancer in only about 20% of cases.

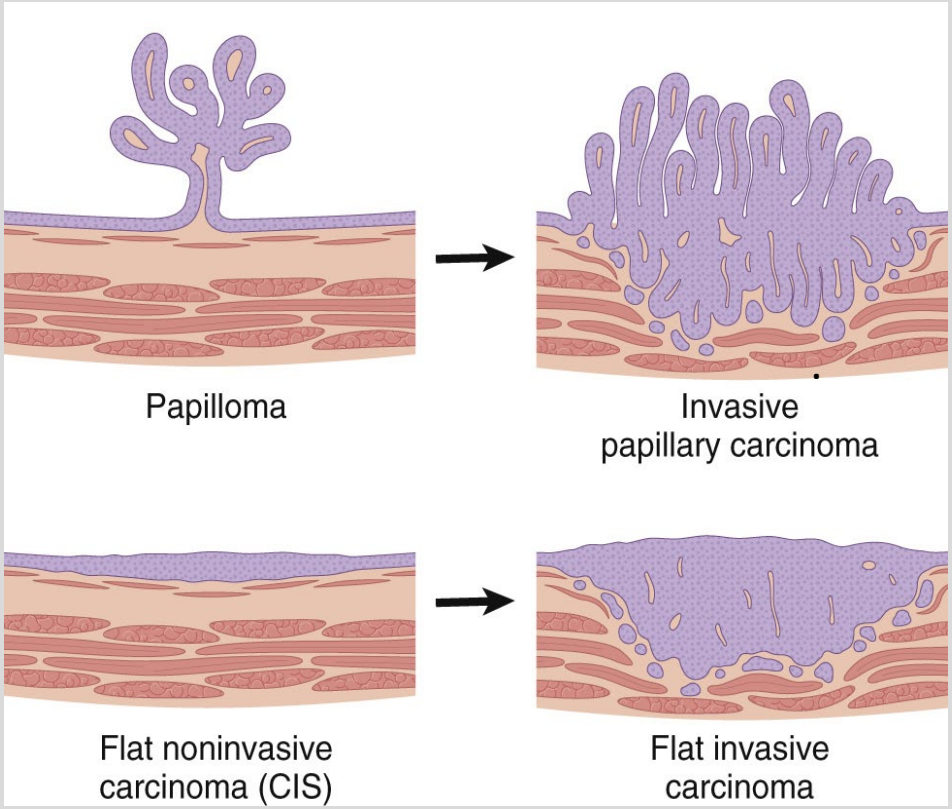
The majority of muscle-invasive bladder cancers develop by progression from "flat" CIS.

- These lesions have mutations of p53 and RB
- These mutations occur early in the development of CIS and later in the progression of papillary cancers.
- high grade as they is multifocal



Urothelial Neoplasms

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



Noninvasive papillary tumors
-most common; may become invasive in some cases

Flat carcinoma in situ CIS.
-epithelial lesion with features of malignancy but confined to the epithelium
-frequently becomes invasive

Urothelial Neoplasms Morphology

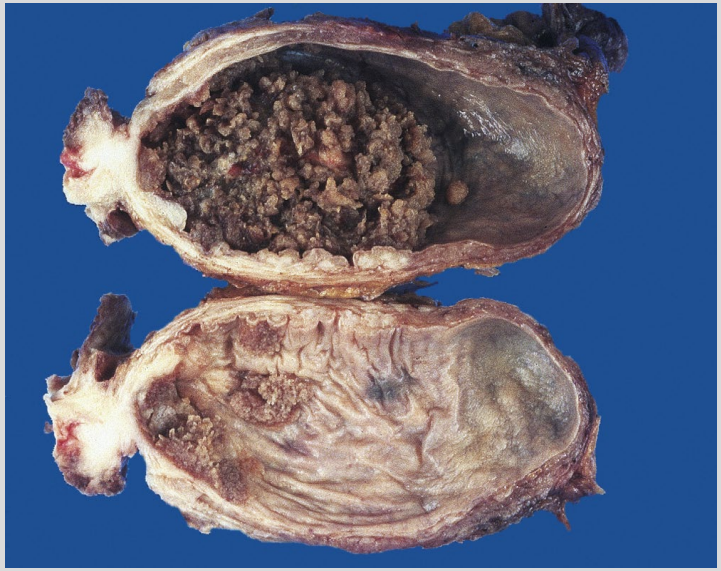
The appearance of urothelial tumors varies from purely papillary, nodular or flat.

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

WHO/ISUP Grades (2016)
Flat Lesions
• Urothelial proliferation of uncertain malignant potential (flat hyperplasia)
• Urothelial dysplasia
• Urothelial carcinoma in situ
Exophytic Papillary Lesions
• Papilloma
• Urothelial proliferation of uncertain malignant potential (papillary hyperplasia)
• Papillary urothelial neoplasms of low malignant potential
• Papillary urothelial carcinoma, low grade
• Papillary urothelial carcinoma, high grade

ISUP, International Society of Urological Pathology; *WHO*, World Health Organization.



The the upper half is filled by a large papillary tumor. The lower half displays multifocal smaller papillary neoplasms.(Courtesy Dr. Fred Gilkey, Sinai Hospital, Baltimore, MD.)

NEW YORK INSTITUTE OF TECHNOLOGY

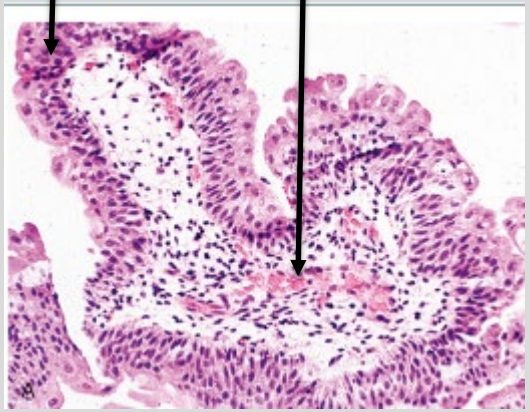
College of Osteopathic Medicine

Papillomas

1% or less of bladder tumors, often seen in younger patients. .

The individual finger-like papillae have a central core of loose fibrovascular tissue covered by epithelium that is histologically identical to normal urothelium.

Recurrences and progression are rare



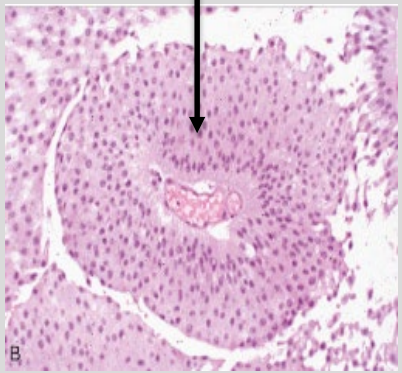
Papilloma consisting of small papillary fronds lined by normal-appearing urothelium.

Papillary urothelial neoplasms of low malignant potential (PUNLMP)

These are vey similar with papillomas, differing only in having thicker urothelium with greater density of cells.

They are larger than papillomas .

Progression to tumors of higher grade may occur but is rare.



Papillary urothelial neoplasms of low malignant potential (PUNLMP) : thicker urothelium with greater density of cells.

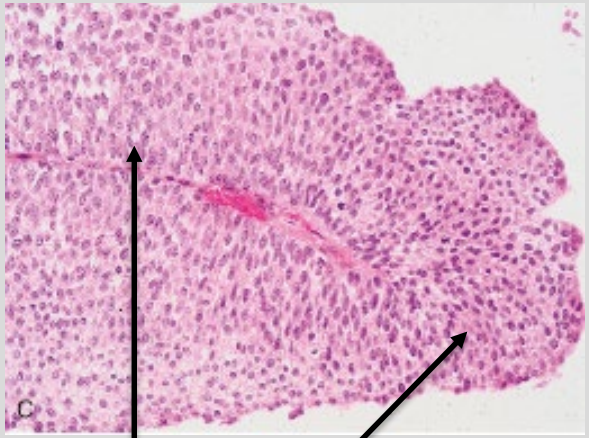
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Low-grade papillary urothelial carcinomas

They have an orderly architectural appearance showing low-grade cytologic atypia.

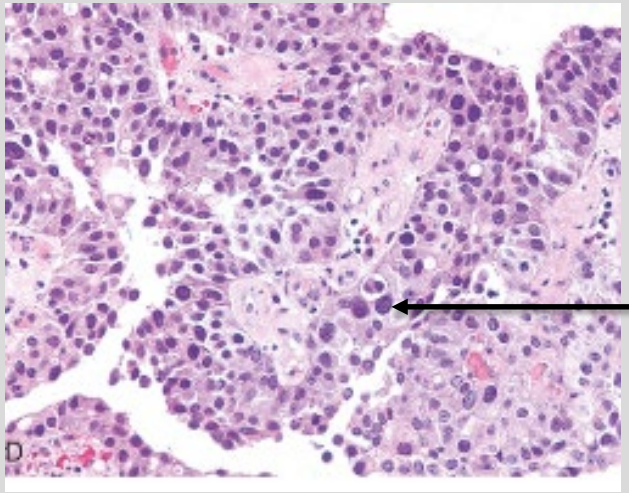
These low-grade cancers may recur but invade invade rarely.



Low-grade papillary urothelial carcinoma with an overall orderly appearance, with scattered hyperchromatic nuclei

High-grade papillary urothelial carcinomas

There is disarray and loss of polarity, and they have a greater incidence of progression to muscle-invasive bladder cancer. Also, they have a significant potential for metastasis to regional lymph nodes and systemic spread to liver and lung.



High-grade papillary urothelial carcinoma with marked cytologic atypia: hyperchromasia and prominent nucleoli.

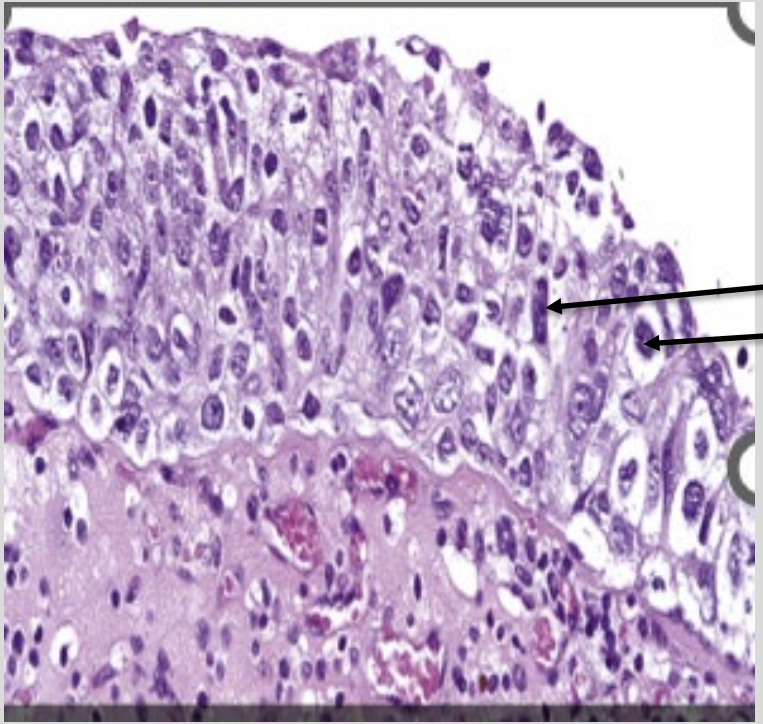
CIS

It is multifocal. If untreated, 50% to 75% of CIS progresses to invasive cancer.

Cystoscopy: an area of mucosal reddening, granularity, or thickening without an evident intraluminal mass.

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



Histo: full-thickness cytologic atypia involving the urothelium (enlarged pleomorphic nuclei):
large cells with pleomorphic nuclei and obvious nucleoli.

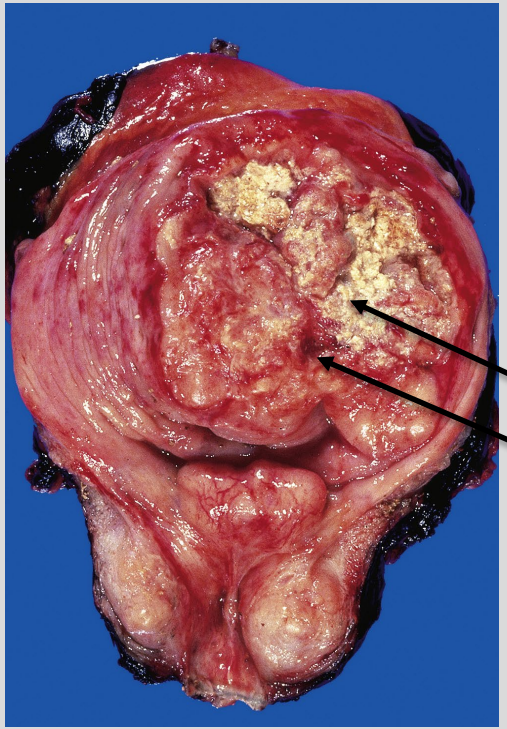
As in the high-grade papillary urothelial carcinoma there is a lack of cohesiveness, which leads to shedding of malignant cells into the urine.

Invasive urothelial carcinoma

The most important prognostic factor is extent of spread-*STAGE*:
it is based primarily on depth of invasion in the bladder wall, at the time of initial diagnosis.

Stage determines treatment modality: invasion of the muscularis propria layer is an indication for radical cystectomy or radiation therapy or adjuvant chemotherapy.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



Opened bladder showing a high-grade invasive transitional cell carcinoma at an advanced stage:
This is an aggressive multinodular neoplasm has fungated into the bladder lumen and spread over a wide area.
Necrosis
Ulceration

Secondary malignant of the bladder

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

: most often due to direct extension of cancers arising in adjacent organs:

- Cervix
- Uterus
- Prostate
- Rectum

Lymphoma may involve the bladder as a component of systemic disease.

Pathology of the Urethra: Inflammation

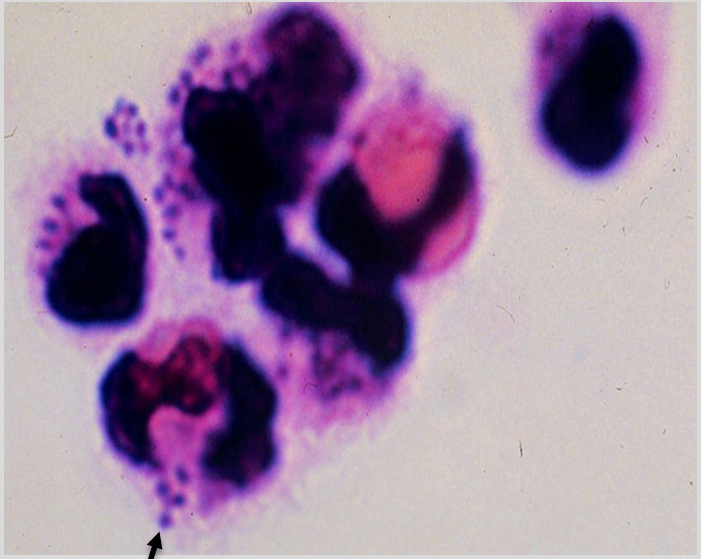
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Gonococcal urethritis is one of the earliest manifestations of this venereal infection.
Nongonococcal urethritis is common and can be caused by several different organisms.

- Various strains of Chlamydia (Chlamydia trachomatis) : 25% to 60% of nongonococcal urethritis in men and about 20% in women.

Urethritis is often accompanied by cystitis in women and by prostatitis in men.

Non-infectious urethritis: In Reactive Arthritis aka Reiter syndrome:
-clinical triad of arthritis, conjunctivitis, and urethritis



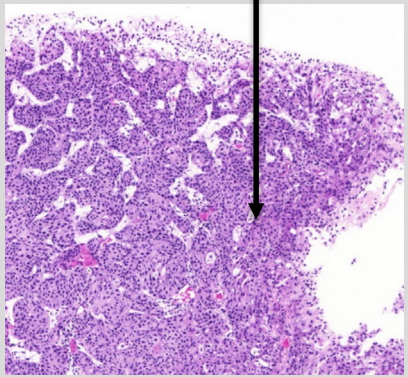
Urethritis discharge: PMNs containing diplococci corresponding to **N. gonorrhoeae.**

Tumor-Like lesions and tumors

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

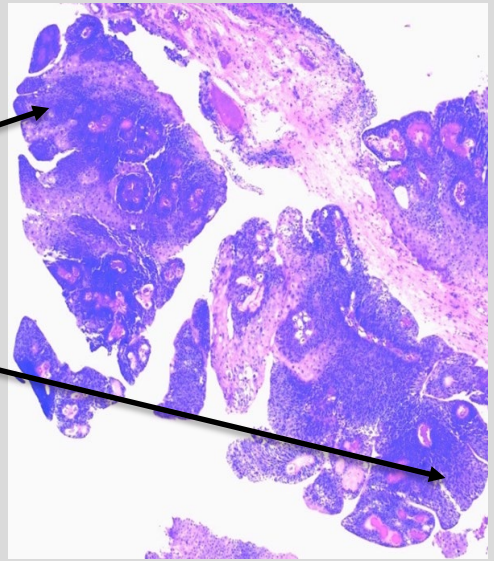
Urethral caruncle is an inflammatory lesion:
-small, red, painful mass about the external urethral meatus, typically in older females.
-inflamed granulation tissue covered by an intact but friable mucosa which ulcerates and bleeds with the slightest trauma.
-surgery provides cure.

Benign epithelial tumors of the urethra include urothelial papillomas, inverted urothelial papillomas, and condylomas.



Inverted urothelial papilloma: Urothelium invaginates into the lamina propria

Condyloma: Polypoid lesion - of hyperplastic papillary fronds of squamous epithelium with koilocytosis-viral damage (not seen at the low power magnification).
It is caused by HPV (both low risk HPV and high-risk HPV).



Primary carcinoma of the urethra

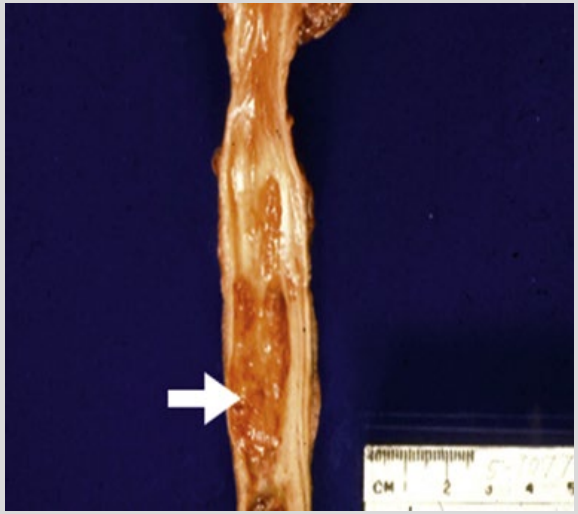
1% or urinary tract malignancies: **Urothelial and Squamous**

Proximal urethra: urothelial differentiation and are analogous to those occurring within the bladder

Distal urethra: squamous cell carcinomas and HPV-related.

Overall, urethral carcinoma is aggressive.
It determines metastasis in: lung, liver, bone.

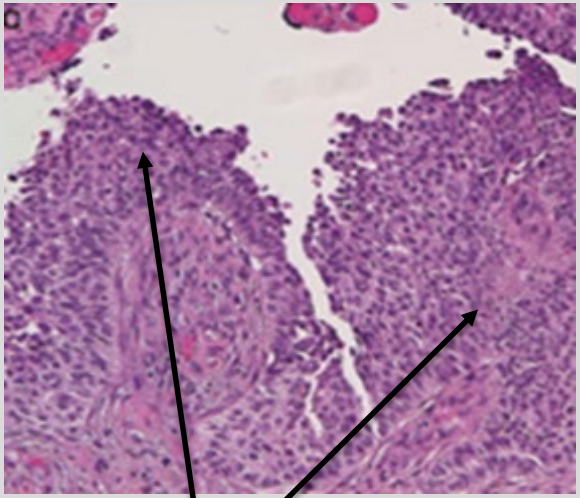
Adenocarcinomas are infrequent in the urethra and generally occur in women



Carcinoma of urethra with a fungating growth pattern (arrow).

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



SCC of the urethra: hyperchromatic malignant cells below basement membrane, with active nuclei and visible nucleoli (at high power), atypical mitosis.

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R